FLea: Addressing Data Scarcity and Label Skew in Federated Learning via Privacy-preserving Feature Augmentation

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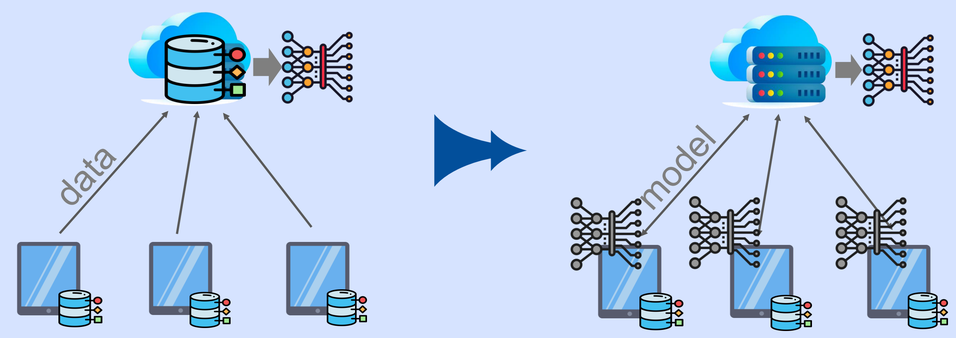
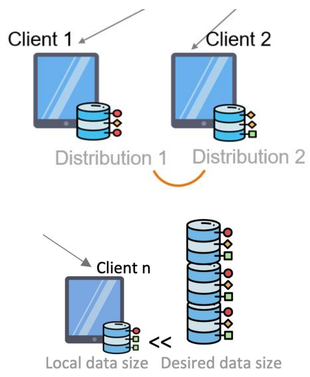
Background



- Traditional DL:
  - Aggregating data in a center server
- Federated Learning (FL):
  - Keeping data on edge devices

Challenges

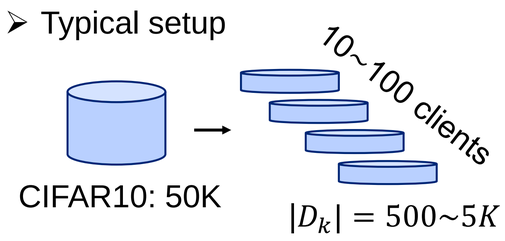
- Data heterogeneity: Label skewness causes local models to drift, biasing them toward local distributions and hindering generalization to the global distribution, resulting in a suboptimal global model [10, 16, 21].
- Local data scarcity: Small datasets can lead to overfitting in local models, limiting their ability to generalize to unseen data and producing a suboptimal global model (***under-explored***).



Problem of data scarcity

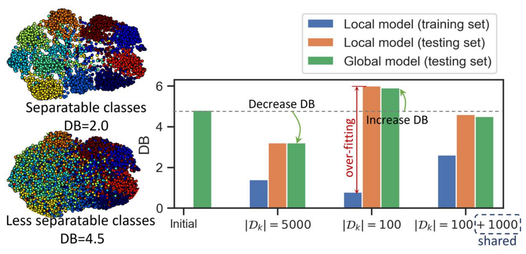
- Existing solutions for label skew
  - Loss-based methods
    - FedDecorr (ICLR'22)
    - FedNTD (NeurIPS'22)
  - Augmentation-based methods
    - FedData (Arkiv'18)
    - FedMix (ICLR'20)

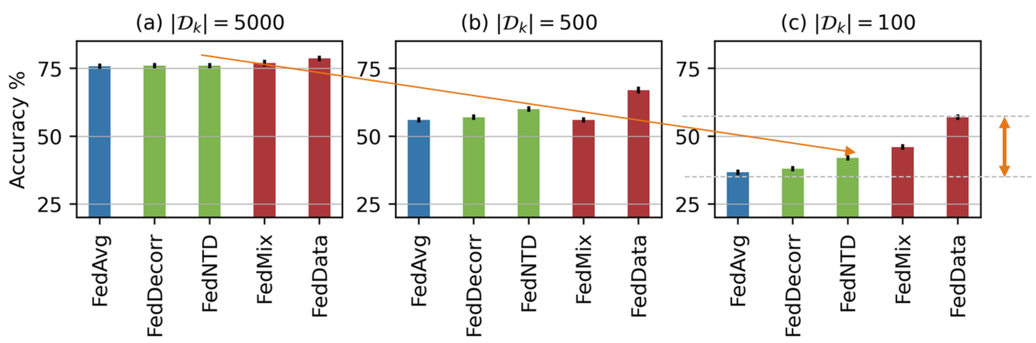
➢ Typical setup



- A controlled IID setup:

How these methods perform when local data are **IID but scarce**?





Data scarcity → local overfitting → sub-optimal global model

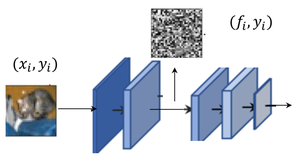
Data sharing → improved generalization

From sharing data to sharing privacy-preserving features

Sharing data

✓ Improved performance

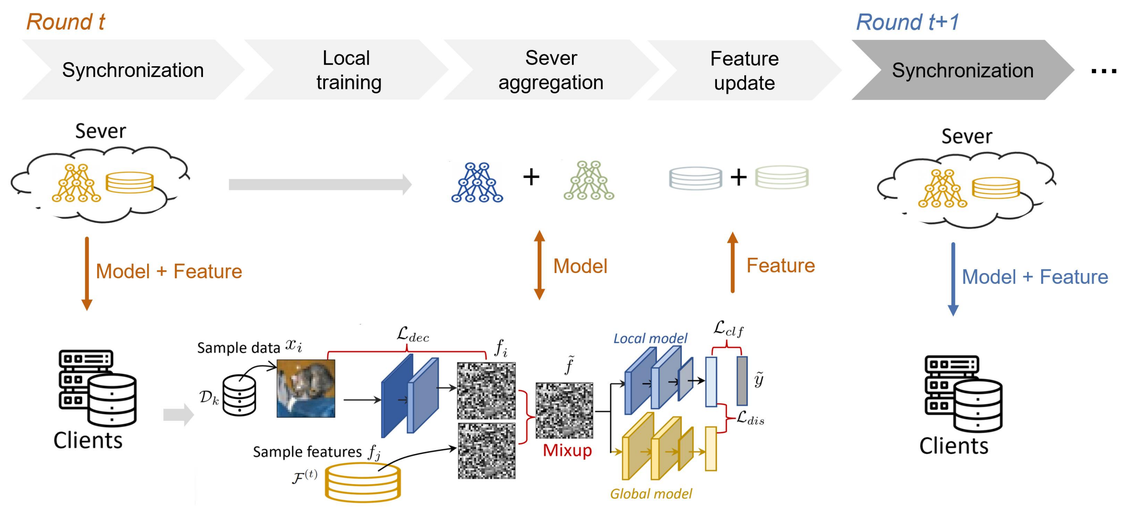
? Breaking privacy



✓ Features: (activation, label)

Research question: Can we generate and share some **privacy-preserving information** as a robust proxy for global data distribution to aid local learning?

Our solution *FLea*

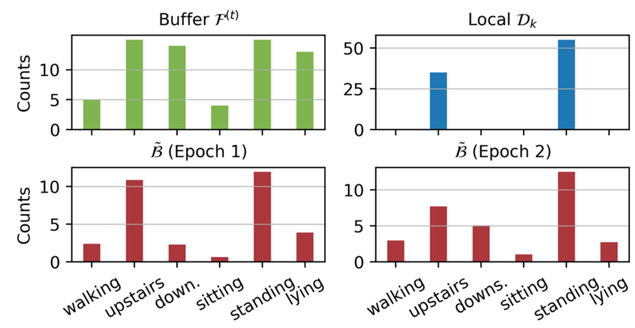


- Global feature buffer:

$$\mathcal{B}^{\mathcal{F}} = \{(f_i^{\mathcal{F}}, y_i^{\mathcal{F}}) \in \mathcal{F}^{(t)}\}$$

- Local feature augmentation:

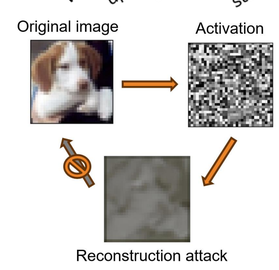
$$\begin{cases} \tilde{f}_i = \beta_i f_i + (1-\beta_i) f_i^{\mathcal{F}}, \\ \tilde{y}_i = \beta_i y_i + (1-\beta_i) y_i^{\mathcal{F}}, \end{cases}$$



- To protect the privacy of the shared features:

$$\mathcal{L}_{dec}(\mathcal{B}) = \frac{v^2(x,f)}{\sqrt{v^2(x,f)v^2(f,f)}},$$

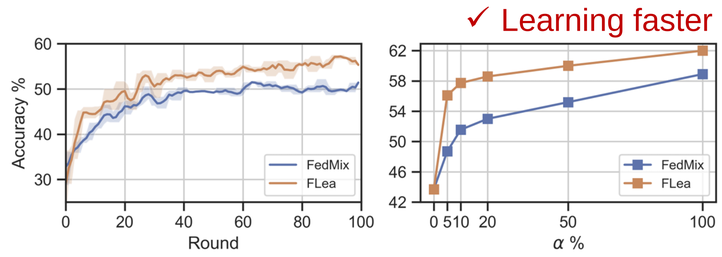
$$\mathcal{L} = \mathcal{L}_{clf}(\mathcal{B}, \mathcal{B}^{\mathcal{F}}) + \lambda_1 \mathcal{L}_{dis}(\mathcal{B}, \mathcal{B}^{\mathcal{F}}) + \lambda_2 \mathcal{L}_{dec}(\mathcal{B}),$$



Experiments and results

- **Datasets**:
  - Image: CIFAR10
  - Audio: UrbanSound8K
  - Time series: UCI-HAR
- **Setup**:
  - Data size: $|D_k| = 50, |D_k| = 100$
  - Label skew: $\mathrm{Qua}(3), Dir(0.1), Dir(0.3)$
- **Baselines**:
  - loss-based methods:
    - FedAvg, FedProx, FedDecorr, FedLC, FedNTD
  - augmentation-based methods:
    - FedBR, CCVR, FedGen, FedMix, FedData

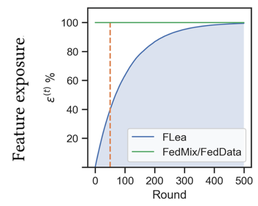
✓ Learning faster



✓ Performing better

Achieving an improvement of over **5%** on **13** out of 18 settings

✓ Privacy preserved



Conclusions

✓ **Contributions**:
- ✓ The first in-depth study for label-skew and data-scared FL
- ✓ A novel feature-sharing FL framework *FLea*
- ✓ Extensive experiments demonstrate the superior performance of FLea

- **Future work**:
  - Theoretical analysis of privacy leakage of shared features
  - Label-free activation-based augmentation improvement
  - Explore FLea for fine-tuning foundation models and LLMs

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Paper

code

Personal web

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